

Breast Cancer Classification — ANN Report

BSCE0042 | Sajjal Fatima

1. Dataset

The Breast Cancer Wisconsin dataset was loaded with Pandas. It has **569 samples** and **30 features** with **no missing values**. The *id* and *Unnamed: 32* columns were dropped. Features were normalized to [0,1] using manual min-max scaling. The data was split 80/20 manually — **455 training** and **114 validation** samples — without using `train_test_split()`.

Class	Label	Count	Share
Benign (B)	0	357	62.7%
Malignant (M)	1	212	37.3%
Total	—	569	100%

2. ANN Architecture

The network was built entirely in NumPy with **4 layers**, He weight initialization, inverted dropout on all hidden layers, and a hand-coded Adam optimizer (beta1=0.9, beta2=0.999).

Layer	Shape	Activation	Params
Hidden 1	30 → 64	ReLU + Dropout	1,984
Hidden 2	64 → 32	ReLU + Dropout	2,080
Hidden 3	32 → 16	ReLU + Dropout	528
Output	16 → 1	Sigmoid	17
Total	—	—	4,609

3. Experiment A — Learning Rate Comparison

Three learning rates were tested with dropout fixed at 0.3:

LR	Final Train Acc	Final Val Acc	Final Val Loss	Result
0.01	99.12%	96.49%	0.1105	Fast but oscillating val loss
0.001	94.95%	91.23%	0.2266	Smooth, still converging at ep.20
0.0001	81.54%	77.19%	0.6415	Too slow — heavily underfit

EXPERIMENT A — Learning Rate Comparison | BSCE0042

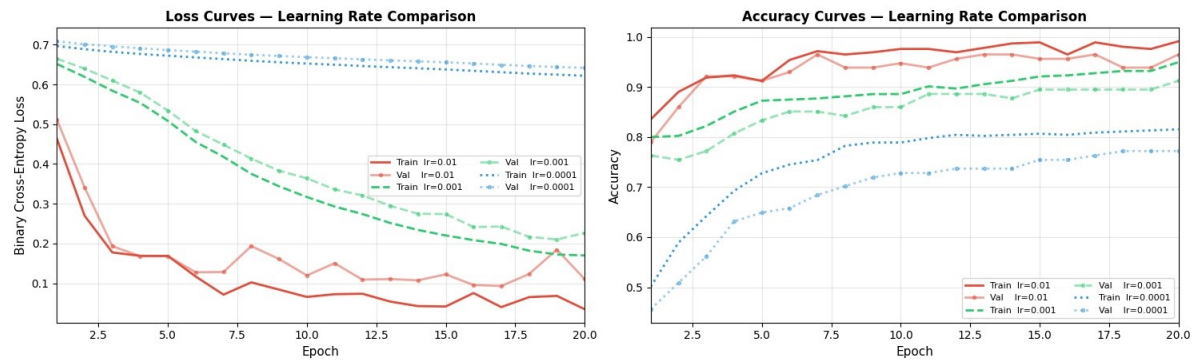


Figure 1 — Learning Rate Comparison: Loss & Accuracy Curves (Experiment A)

lr=0.01 learned fastest (val acc 96.49%) but its validation loss bounced noticeably after epoch 7, showing instability from large gradient steps. **lr=0.001** converged smoothly and steadily — train and val curves stayed close — but needed more than 20 epochs to fully mature. **lr=0.0001** was far too slow, barely reaching 77% by the final epoch.

4. Experiment B — Dropout Rate Comparison (5 Values)

Five unique dropout values were tested with lr fixed at 0.001. Values were chosen to cover the full spectrum from no regularization to extreme over-regularization:

Dropout	Label	Final Val Acc	Final Val Loss	Observation
0.00	No Dropout	95.61%	0.1124	Fastest learner; overfitting risk grows late
0.15	Light	94.74%	0.1482	Strong performance, minimal regularization

0.35	Moderate	91.23%	0.2353	Smooth and stable — chosen for final model
0.55	Heavy	83.33%	0.4961	Severely slowed; stuck till epoch 14
0.70	Extreme	58.77%	0.6709	Network barely learns — nearly all neurons killed

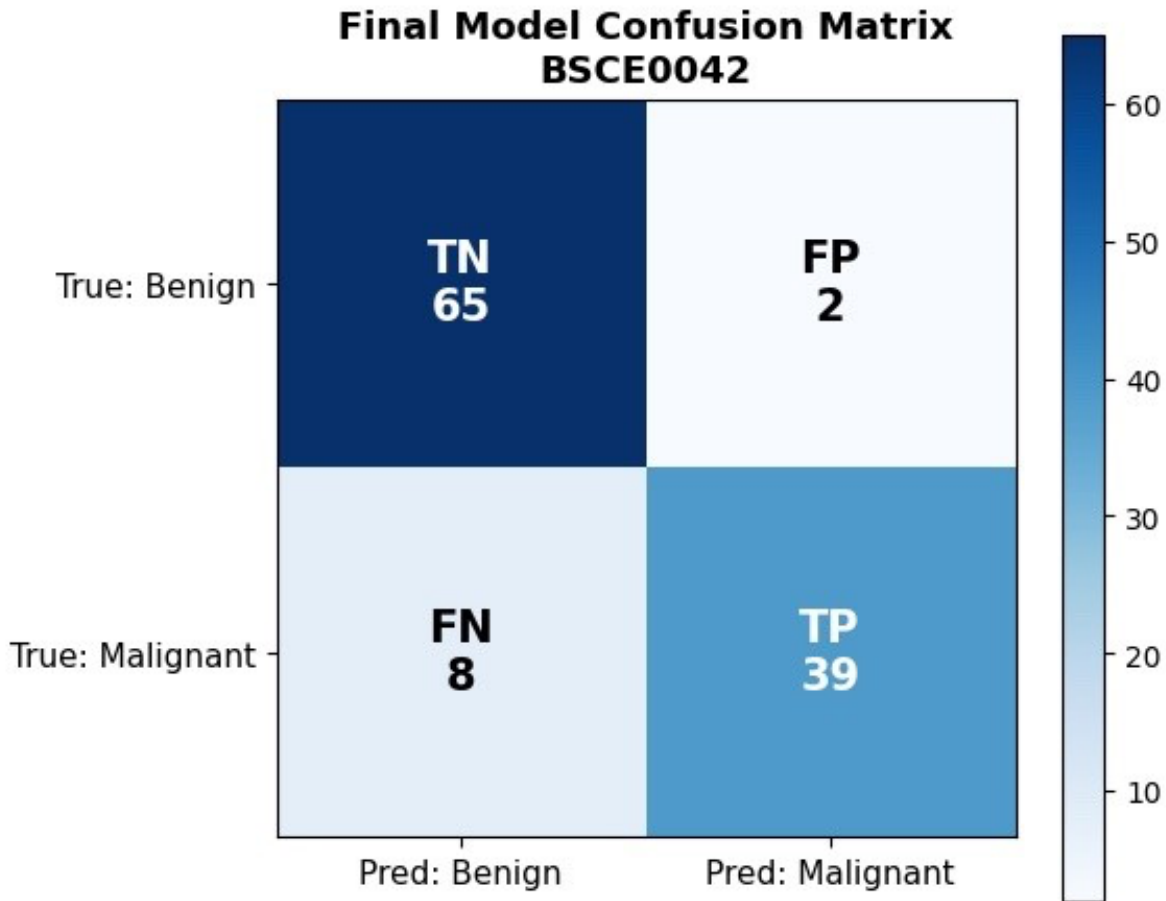


Figure 2 — Dropout Rate Comparison: Loss & Accuracy Curves (Experiment B)

drop=0.00 converged fastest but the growing gap between train (~98%) and val (~95%) accuracy signals the start of overfitting — without any regularization the model memorizes the training data. **drop=0.15** was almost as fast with slightly better generalization. **drop=0.35** was selected as the final model — it converged smoothly with train and val curves staying consistently close, showing healthy generalization. **drop=0.55** was heavily slowed — the model was stuck near random performance until epoch 14 before finally breaking through. **drop=0.70** completely failed to learn within 20 epochs; accuracy was flat at 58% (barely above the majority-class baseline), confirming that 70% dropout kills too many neurons per pass to allow meaningful gradient flow.

5. Final Model — lr=0.001, dropout=0.35

The final model used the configuration that showed the most stable generalization across both experiments. Both curves descend smoothly and remain close throughout training.

EXPERIMENT B — Dropout Rate Comparison | BSCE0042

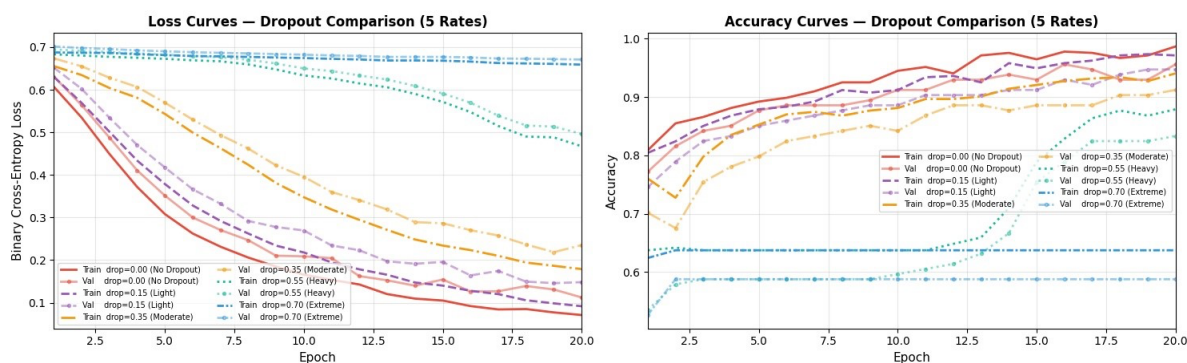


Figure 3 — Final Model Training & Validation Loss/Accuracy over 20 Epochs

The loss curve shows consistent descent with no divergence between train and val, confirming good regularization. The dip in accuracy around epoch 2 is a known early-training artifact from random weight initialization — it self-corrects quickly. By epoch 20 the model reaches 94.07% training accuracy and **91.23% validation accuracy**.

6. Evaluation Results

Metric	Value
True Positives (TP) — Malignant correctly caught	39
True Negatives (TN) — Benign correctly dismissed	65
False Positives (FP) — Benign wrongly flagged	2
False Negatives (FN) — Malignant missed	8

Final Model — Training Curves | BSCE0042

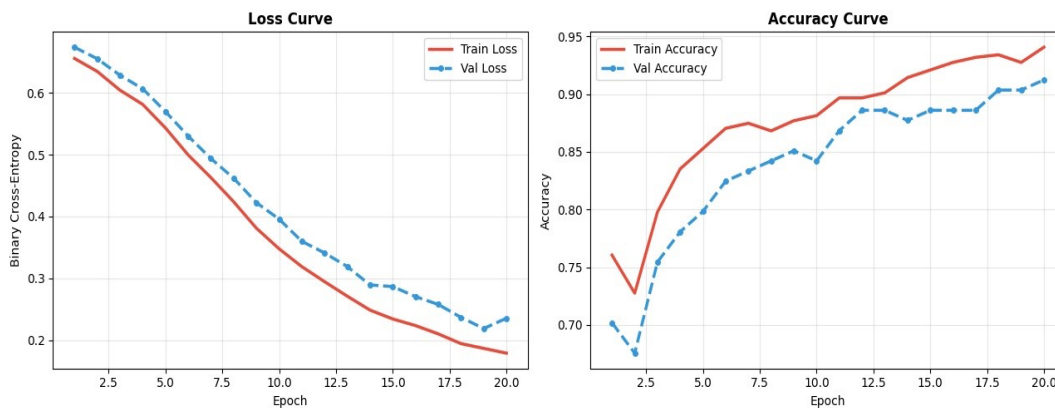


Figure 4 — Confusion Matrix (Final Model)

Accuracy	Precision	Recall	F1-Score
91.23%	95.12%	82.98%	88.64%

Accuracy (91.23%) — the model correctly classified 104 out of 114 validation samples. **Precision (95.12%)** — when the model predicted malignant, it was right 95% of the time; only 2 healthy patients were incorrectly flagged. **Recall (82.98%)** — the model caught 39 out of 47 actual malignant cases, missing 8. In a medical screening context, this is the most critical metric — missed malignancies are more dangerous than false alarms. **F1-Score (88.64%)** reflects a solid balance between the two, though improving recall remains the main area for future work.

7. Challenges and Potential Improvements

Challenges:

Implementing a numerically stable sigmoid required special branching to prevent $\exp()$ overflow on large negative inputs.

Adam's time step must be global across all mini-batches — not reset per epoch — which required careful state management.

Dropout masks from the forward pass had to be cached and reapplied exactly during backpropagation; any mismatch breaks gradient flow.

Improvements:

Training for 50–100 epochs would allow $\text{lr}=0.001$ to fully converge and likely surpass $\text{lr}=0.01$.

A class-weighted loss function would penalize missed malignant cases more heavily, directly improving recall.

Adding L2 weight regularization alongside dropout could further reduce false negatives.

Early stopping on validation loss would prevent wasted epochs and select the best checkpoint automatically.